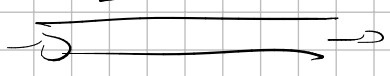
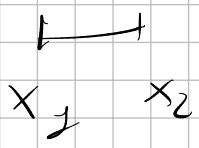
$$\rho(t,x) = \text{"}\frac{\#\ \text{particelle}}{\text{volume}}\text{"} = \frac{\text{massa del gas}}{\text{volume del gas}}$$

1D

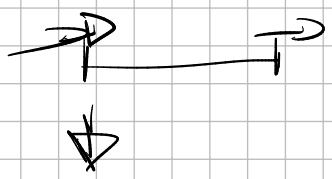




$$\int_{x_1}^{x_2} \rho(t,x)\,dx$$ = MASSA DEL NOSTRO GAS in $[x_1,x_2]$ AL VARIARE DEL TEMPO

$v(t,x)$ velocità del gas

MASSA SI CONSERVA A MENO DI EFFETTI AL BORDO



FLUSSO

$$\underbrace{v(t,x_2)\cdot\rho(t,x_2)}$$

NEL TEMPO $[t_1,t_2]$

VARIAZIONE IN TEMPO DI MASSA

$$\int_{x_1}^{x_2} \rho(t_2,x) - \rho(t_1,x)\,dx =$$

$$= -\int_{t_1}^{t_2} \rho(t,x_2)\,v(t,x_2) - \rho(t,x_1)\,v(t,x_1)\,dt$$

QUANTA MASSA È ENTRATA / USCITA CON SEGNO AI BORDI

$$\frac{d}{dt}\int_{x_1}^{x_2} \rho(t,x)\,dx = -\rho(t,x_2)\cdot v(x_2,t) + \rho(t,x_1)\,v(x_1)$$

1) $\frac{d}{dt}\rho(t,x) = -\frac{\partial}{\partial x}\underbrace{\left(\rho(t,x)\, v(t,x)\right)}_{f(\rho(t,x))}$?

CONSERVAZ. MASSA

$$\boxed{\frac{d}{dt}\rho(t,x) + \partial_x f(\rho(x,t)) = 0}$$

CONSERVAZIONE MOMENTO

2) $\partial_t(\rho v) + \partial_x(\rho v^2 + p) = 0$

ORA SAPPIAMO LA VELOCITÀ

?

$p(\rho)$

CONSERVAZIONE DELL'ENERGIA

3) $\partial_t(\rho E) + \partial_x(\rho v E + v \cdot p) = 0$

$p \Leftrightarrow E$ EQUAZIONE DI STATO

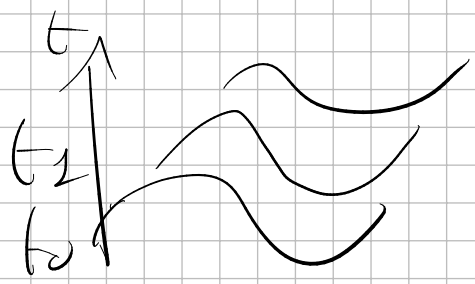
$p = (\gamma - 1)\cdot\rho\left(E - \frac{v^2}{2}\right)$ γ COSTANTE ADIABATICA

$E = \frac{p}{\rho(\gamma-1)} + \frac{v^2}{2}$ $\gamma = \frac{7}{5}$

TRASPORTO LINEARE

$\partial_t u(t,x) + a\,\partial_x u(t,x) = 0$ $a \in \mathbb{R}$

$u(t,x) = u_0(x - at)$



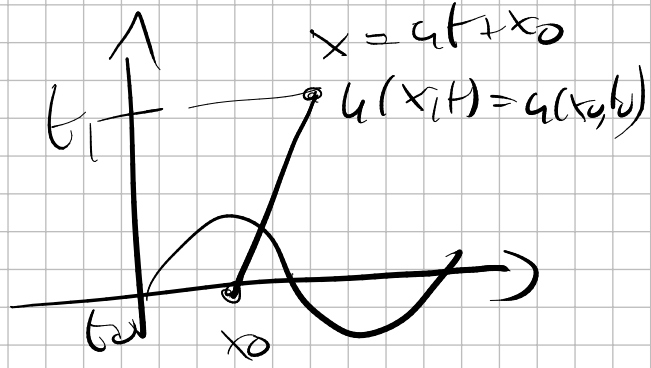
• CARATTERISTICHE

$$x = at + x_0 = x(t)$$

$$0 = \frac{d\,u(t, x(t))}{dt} = \partial_t u(t, x(t)) + \partial_x u(t, x(t)) \cdot \underbrace{\partial_t x(t)}$$

$$\Rightarrow \quad \partial_t x(t) = -\frac{\partial_t u}{\partial_x u} = a \quad = \text{costante}$$

$$x(t) = x_0 + at$$



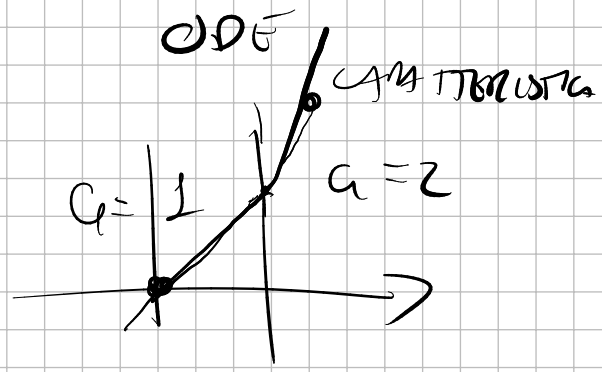
$$\Downarrow$$

$$u_t + a(x) \cdot u_x = 0$$

$$\partial_t x(t) = a(x(t)) \Rightarrow \text{RISOLVERE ODE}$$

$\Downarrow$

CARATTERISTICA



NON È UNA LEGGE DI CONSERVAZIONE

$$\int_{x_1}^{x_2} \partial_t u + \underbrace{\int_{x_1}^{x_2} a(x) \cdot u_x} = 0$$

$\Downarrow$

A PRIORI NON È UNA DIFFERENZA DI FLUSSI

$$u_t + (a(x)\cdot u)_x = 0$$

LO È UNA LEGGE DI CONSERVAZIONE!

$$\underbrace{u_t + a(x)\cdot u_x}_{\text{cerchiamo caratteristiche di questa parte}} + \underbrace{a_x(x)\cdot u(x)}_{\text{non facciamo ODE}} = 0$$

$$\partial_t X(t) = a(x) \quad \forall x_0 \longrightarrow \text{ODE \#1}$$

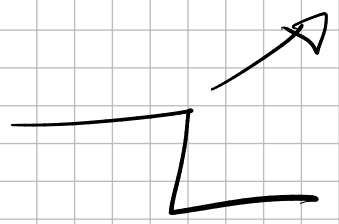
CARATTERISTICHE

$$\frac{d\,u(t,x(t))}{dt} = -a_x(x(t))\cdot u(x(t))$$

ODE #2

$u \in L^2(\mathbb{R}^+ \times \Omega)$

$[x_1, x_2] = \Omega \subseteq \mathbb{R}$



$$u_t + \partial_x(a(x)\cdot u(t,x)) = 0 \qquad \varphi \in C^\infty(\mathbb{R}^+ \times \Omega)$$

$$\int_0^T \int_\Omega \varphi(x,t)\cdot u_t(x,t) + \varphi(x,t)\,(a(x)\cdot u(t,x))_x = 0$$

INTEGR. PER PARTI IN t

I.P.P.

$$\int_\Omega \varphi(x,T)\, u(x,T) - \varphi(x,0)\, u(x,0)\, dx$$

$$- \int_0^T \int_\Omega \partial_t \varphi(x,t) \cdot u(x,t)\, dt\, dx + \int_0^T \varphi(x_R,t)\, a(x_R)\, u(x_R,t)\, dt$$

$$- \int_0^T \varphi(x_L,t)\, a(x_L)\, u(x_L,t)\, dt$$

$$- \int_0^T \int_\Omega \partial_x \varphi(x,t) \cdot a(x)\, u(x,t)\, dx\, dt = 0$$

HA DEFINITAMENTE SENSO PER $u \in L^2$

SOL DI PDE DEBOLE È TROVA $a \in L^2$:

$$\forall \varphi \in C^\infty$$

Se scelgo $\varphi = u$ $\qquad \frac{u^2}{2}$ = "ENERGIA"

$$\int_{t_0}^{t_1} \int_{x_L}^{x_R} u \cdot \partial_t u + \int_{t_0}^{t_1} \int_{x_L}^{x_R} u\, \partial_x (a \cdot u(x,t))\, dx\, dt = 0$$

$$\int_{t_0}^{t_1} \underbrace{\int_{x_L}^{x_R} \partial_t \left(\frac{u^2}{2}\right)} + \int_{t_0}^{t_1} \int_{x_L}^{x_R} \partial_x \left(a \frac{u^2}{2}\right) dx\, dt = 0$$

$$\int_{x_L}^{x_R} \frac{u^2(t_1)}{2} - \frac{u^2(t_0)}{2}\, dx + \int_{t_0}^{t_1} a \frac{u^2(x_R)}{2} - a \frac{u(x_L)^2}{2}\, dt = 0$$

CONDIZ. BORDO ONO DIRI / PERIODICHE

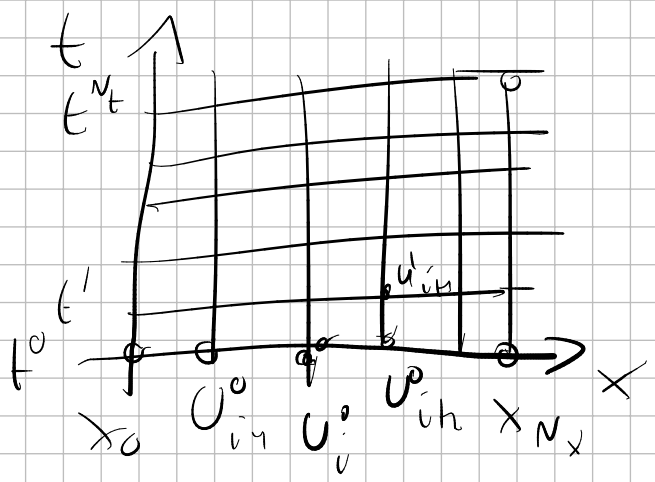
$$\int_{x_L}^{x_R} \frac{u^2(t_1)}{2} = \int_{x_L}^{x_R} \frac{u^2(t_0)}{2}$$

ENERGIA SI CONSERVA (A MENO DI CONTRIBUTI DAI BORDI)

$$\partial_t u^\varepsilon + a\,\underbrace{\partial_x u^\varepsilon} = \underbrace{\varepsilon\, u^\varepsilon_{xx}}_{\text{VISCOSITÀ}}$$

$$\lim_{\varepsilon \to 0} u^\varepsilon(t,x) = \underbrace{u(t,x)}_{\text{SOLUZIONI A VISCOSITÀ EVANESCENTE}}$$

$$\partial_t u + \partial_x u = 0 \qquad a = 1$$



$$\underbrace{(\partial_x u)_i \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}}$$

$$\approx \frac{1}{\Delta x}$$

$$u_i^n \approx u(t^n, x_i)$$

EULERO ESPLICITO (STABILE) $\Delta t \sim \Delta x$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

È STABILE? STABILITÀ DI VON NEUMANN

$$u_j^n = e^{i \underbrace{j\Delta x}_{x} \cdot k} \cdot c_k^n \quad k \in \mathbb{Z}$$

$$c_k^{n+1} \cdot \underbrace{e^{i j \Delta x k}} = c_k^n \underbrace{e^{i j \Delta x k}} - \frac{\Delta t}{2\Delta x} \cdot$$

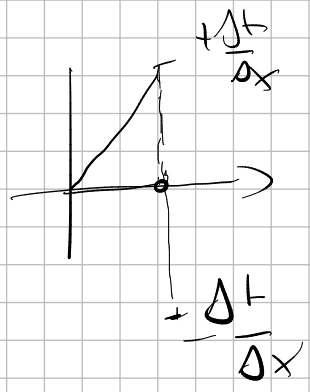
$$\cdot \left(e^{i (j+1) \Delta x k} - e^{i (j-1) \Delta x k} \right) c_k^n$$

$$c_k^{n+1} = c_k^n \underbrace{\left(1 - i \frac{\Delta t}{\Delta x} \boxed{\left(\frac{e^{i \Delta x k} - e^{-i \Delta x k}}{2i} \right)} \right)}_{g(k)}$$

COEFFICIENTE AMPLIFICAZ.

$$|g(k)| \leq 1 + \alpha \Delta t \qquad \alpha \text{ INDIP. DA } \Delta t$$

$$g(k) = 1 - i \frac{\Delta t}{\Delta x} \sin(k \Delta x)$$



$$1 \leq |g(k)|^2 \leq 1 + \frac{\Delta t^2}{\Delta x^2}$$

$$\Delta t \simeq \Delta x^2$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + \frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} = 0$$

$$c_k^{n+1}\left(\frac{1}{\Delta t} + \frac{1}{\Delta x}\underbrace{\left(\frac{e^{ik\Delta x} - e^{-ik\Delta x}}{2}\right)}_{i\sin(k\Delta x)}\right) = c_k^n \frac{1}{\Delta t}$$

$$c_k^{n+1} = \frac{1}{1 + \frac{\Delta t}{\Delta x} i\sin(k\Delta x)} c_k^n$$

$$|g(k)| = \underbrace{\left|1 + i\frac{\Delta t}{\Delta x}\sin(k\Delta x)\right|}_{\geq 1}^{-1} \leq 1$$

$$u' = -\lambda u$$

LAX-FRIEDRICHS

$$u_j^{n+1} = \frac{u_{j+1}^n + u_{j-1}^n}{2} - a\frac{\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n)$$

$$= u_j^n + \underbrace{\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{2}}_{\simeq \Delta x^2 u_{xx} \simeq \varepsilon \cdot u_{xx}} - a\frac{\Delta t}{2\Delta x}(u_{j+1}^n - u_{j-1}^n)$$

$$g(k) = 1 + \underbrace{\frac{e^{ik\Delta x} - 2 + e^{-ik\Delta x}}{2}} - ia\frac{\Delta t}{\Delta x}\sin(k\Delta x)$$

$$= \cancel{1} + \underbrace{\cos(k\Delta x) - \cancel{1} - i\frac{a\Delta t}{\Delta x}\sin(k\Delta x)}$$

$$= \cos(k\Delta x) - i\frac{a\Delta t}{\Delta x}\sin(k\Delta x)$$

$$|g(k)|^2 = \cos(k\Delta x)^2 + \left(\frac{a\Delta t}{\Delta x}\right)^2 \sin(k\Delta x)^2$$

se $\left(\frac{a\Delta t}{\Delta x}\right)^2 \leq 1$

$\left(|a+ib| = a^2+b^2\right)$

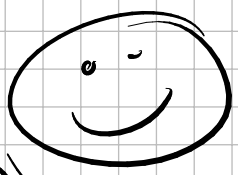
$\Downarrow$

$|g(k)|^2 \leq \cos^2 + \sin^2 = 1$

$$\left|a\frac{\Delta t}{\Delta x}\right| \leq 1 \qquad \Delta t \leq \frac{\Delta x}{|a|}$$

$\Downarrow$

STABILITÀ PER $\Delta t \sim \Delta x$



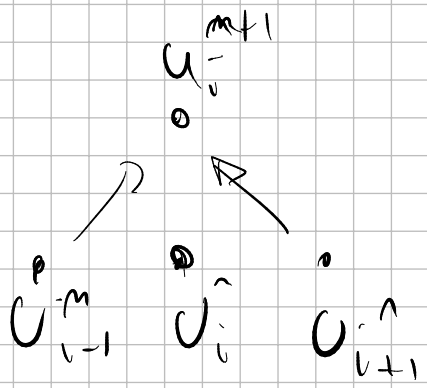
$$\Delta t \leq \min_x\left(\frac{\Delta x}{|a(x)|}\right) = \frac{\Delta x}{\max|a(x)|}$$

$$\Delta t \le \frac{\Delta x}{|a|} \Leftrightarrow CFL$$

$$\Delta t = CFL \cdot \frac{\Delta x}{\max_x |a(x)|}$$

$CFL \le 1$

$CFL \cong 0.9$



$$\Delta t = \frac{\Delta x}{|a|} \qquad \Delta t = \underbrace{CFL}_{\le 1} \frac{\Delta x}{\max_x |a(x)|}$$

$$\left| a \frac{\Delta t}{\Delta x} \right| \le 1 \qquad CFL = 0.9$$

$$u_i^{n+1} = u_i^n + \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{2} - a \frac{\Delta t}{2\Delta x} (u_{i+1}^n - u_{i-1}^n)$$

$$\frac{u(t^n + \Delta t, x_i) - u(t^n, x_i)}{\Delta t} + \frac{u(t^n, x_i + \Delta x) - 2 \cdot u(t^n, x_i) + u(t^n, x_i - \Delta x)}{2\Delta t}$$

$$- a \frac{u(t^n, x_i + \Delta x) - u(t^n, x_i - \Delta x)}{2\Delta x} =$$

$$= \partial_t u + O(\Delta t) +$$

$$\cancel{u} + \cancel{\partial_x u\,\Delta x} + \underbrace{\frac{\Delta x^2}{2}\partial_{xx}u} + \cancel{\frac{\Delta x^3}{6}\partial_{xxx}u} + O(\Delta x^4)$$

$$-\frac{\cancel{-2u} + \cancel{u} - \cancel{\Delta x\,\partial_x u} + \underbrace{\frac{\Delta x^2}{2}\partial_{xx}u} - \cancel{\frac{\Delta x^3}{6}\partial_{xxx}u} + O(\Delta x^4)}{2}$$

$$+a\,\frac{\cancel{u} + \partial_x u\cdot\Delta x + \cancel{\frac{\Delta x^2}{2}\partial_{xx}u} + \frac{\Delta x^3}{6}\partial_{xxx}u - \left(\cancel{u} - \Delta x\,\partial_x u + \cancel{\frac{\Delta x^2}{2}\partial_{xx}u} - \frac{\Delta x^3}{6}\partial_{xxx}u\right)}{2\Delta x}$$

$$= \partial_t u + O(\Delta t) - \frac{\Delta x^2}{2\Delta t}\partial_{xx}u + a\cdot\partial_x u + O(\Delta x^2)$$

$$\partial_t u + a\,\partial_x u - \underbrace{\varepsilon}_{\to 0}\,\partial_{xx}u = O(\Delta x^2) \qquad \Delta t \sim \Delta x$$

$$\underbrace{\Delta x}_{\to 0}\cdot\partial_{xx}u$$

$$a(x)$$

$$\Rightarrow \boxed{\partial_t u + a\,\partial_x u = 0}$$

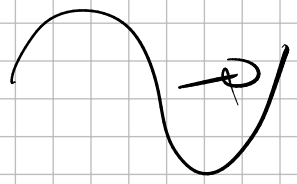
$$u_i^{n+1} - u_i^n = \dots \quad a_i\,\frac{u_{i+1} - u_{i-1}}{2\Delta x}$$

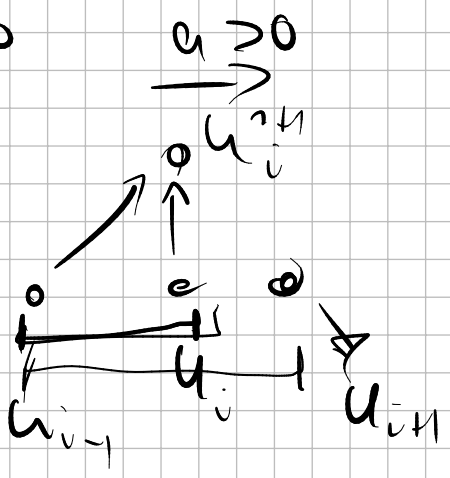
$$a(x_i)$$

$$\partial_x(a(x)\cdot u)$$

$$\frac{a_{i+1}u_{i+1} - a_{i-1}u_{i-1}}{2\Delta x}$$

$$\partial_t u + a\,\partial_x u = 0$$





UPWIND

$$u_i^{n+1} = u_i^n - a\,\Delta t \begin{cases} \dfrac{u_i - u_{i-1}}{\Delta x} & a > 0 \\[2ex] \dfrac{u_{i+1} - u_i}{\Delta x} & a < 0 \end{cases}$$

$a > 0$

$$u_i^{n+1} = u_i^n - a\,\Delta t\,\frac{u_i - u_{i-1}}{\Delta x} =$$

$$= u_i^n - a\,\Delta t\,\frac{u_{i+1} - u_{i-1}}{2\Delta x} + \underbrace{a\,\Delta t\,\frac{u_{i+1} - u_{i-1}}{2\Delta x} - a\,\Delta t\,\frac{u_i - u_{i-1}}{\Delta x}}$$

$$= \underbrace{u_i^n - a\,\Delta t\,\frac{u_{i+1} - u_{i-1}}{2\Delta x}}_{\text{DISCRETIZZ. CENTRATA}} + \underbrace{\frac{a\,\Delta t}{\Delta x}}_{\text{IN LAX-FRIEDRICHS } = 1}\,\frac{u_{i+1} - 2u_i + u_{i-1}}{2}$$

$$a\,\Delta x\,\Delta t\; u_{xx}$$

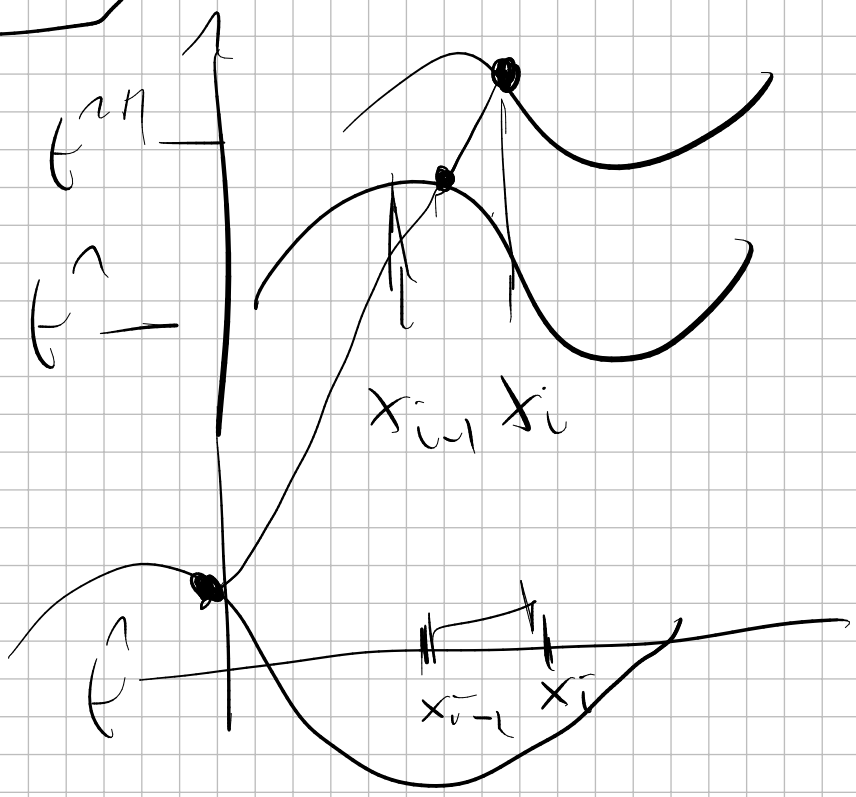
+ DIFFUSIONE ARTIFICIALE

$$-a\frac{\Delta t}{\Delta x}\underbrace{\left(\frac{e^{ik\Delta x}-e^{-ik\Delta x}}{2}\right)}_{i\sin(k\Delta x)}+a\frac{\Delta t}{\Delta x}\left(-1+\underbrace{\frac{e^{ik\Delta x}-e^{-ik\Delta x}}{2}}_{\cos(k\Delta x)}\right)$$

$$-a\frac{\Delta t}{\Delta x}\left[i\sin(k\Delta x)+(\cos(k\Delta x)-1)\right]$$

$$\left|1-a\frac{\Delta t}{\Delta x}\left[i\sin \quad - \quad + \quad - \quad 1\right]\right|^2\leq 1$$

$$\boxed{\left|a\frac{\Delta t}{\Delta x}\right|\leq 1}$$



$$\frac{u_i^{n+1}-u_i^n}{\Delta t}+a\frac{u_{i+1}^n-u_{i-1}^n}{2\Delta x}-\varepsilon\frac{u_{i+1}^n-2u_i^n+u_{i-1}^n}{2\Delta x^2}=0$$

$$\varepsilon = \begin{cases} a\dfrac{\Delta x^2}{\Delta t} \\ a \cdot \Delta x \end{cases}$$

UPWIND

$\varepsilon = C \cdot \Delta x$

$C = \dfrac{\varepsilon}{\Delta x}$

LAX-FRIEDRICHS

1° ORDINE

$$\partial_t u(t,x) + \frac{\Delta t}{2}\partial_{tt} u(t,x) + a\,\partial_x u + a\frac{\Delta x^2}{6}\partial_{xxx} u - \frac{C}{2}\Delta x\,\partial_{xx} u = O(\Delta x^2)$$

$\Delta t \sim \Delta x$

DUBBIO FACCIAMO SPARIRE

$$\frac{\Delta t}{2}\partial_{tt} u - \frac{C}{2}\Delta x\,\partial_{xx} u \to 0 \quad ?$$

2° ORDINE ?

$\parallel$

0

a COSTANTE

- $\partial_t u + a\,\partial_x u = 0$

$\partial_t u = -a\,\partial_x u$

$$\partial_{tt} u = \partial_t(\partial_t u)$$

$\parallel$

$$\partial_t(-a\,\partial_x u) = -a\,\partial_{tx} u$$

$$= -a\,\partial_x(\partial_t u) = -a\,\partial_x(-a\,\partial_x u)$$

$$= \underbrace{a^2\,\partial_{xx} u}$$

$$\frac{\Delta t}{2} a^2\,\partial_{xx} u - \frac{C}{2}\Delta x\,\partial_{xx} u = 0$$

PER AVERE SECOND'ORDINE SCEGLIAMO C

$$\underbrace{\left(\frac{\Delta t}{2}a^2 - \frac{c}{2}\Delta x\right)}\underbrace{\partial_{xx}u}_{\neq 0} = 0$$

$$c = \frac{\Delta t\, a^2}{\Delta x} \Rightarrow \text{SECONDO ORDINE} \ddot{\smile}$$

LAX-WENDROFF MODO 1 $\qquad \partial_{tt}u = a^2\partial_{xx}u$

MODO ALTERNATIVO $\qquad \partial_t u = -a\,\partial_x u$

$$u_i^{n+1} \simeq u_i^n + \Delta t\,\partial_t u_i^n + \frac{\Delta t^2}{2}\partial_{tt}u_i^n + O(\Delta t^3)$$

$$= u_i^n - a\Delta t(\partial_x u)_i^n + \frac{\Delta t^2}{2}a^2(\partial_{xx}u)_i^n + O(\Delta t^3)$$

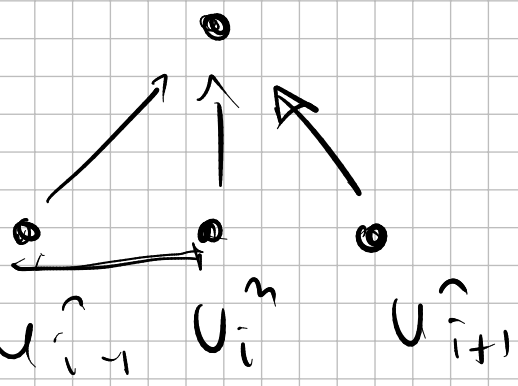
$$= u_i^n - a\Delta t\,\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} + O(\Delta t\,\Delta x^2) + \frac{\Delta t^2}{2}a^2\,\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2} + O(\Delta t^2\Delta x^2) + O(\Delta t^3)$$

$$\approx u_i^n \underbrace{- a\Delta t\,\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} + \frac{\Delta t^2}{2}\frac{a^2}{\Delta x^2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)}$$

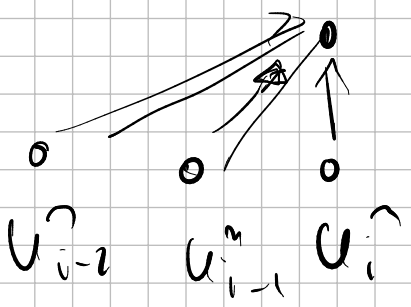
$$\frac{a\Delta t}{\Delta x} = \underbrace{CFL}$$

$$-CFL \cdot \overbrace{i\sin(\theta)} + \underbrace{CFL^2} \cdot \underbrace{(\cos(\theta)-1)}$$

$$\theta = k\Delta x$$



<u>BEAM-WARMING</u> $a > 0$



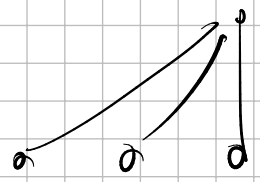
2 OBIETTIVI

1. ALTO ORDINE
2. ESSERE UN PO' PIÙ STABILE

$$\partial_t u_i = -a \underbrace{\frac{u_{i-2} - 4u_{i-1} + 3u_i}{2\Delta x}}_{\partial_x u + O(\Delta x^2)}$$

$$\underbrace{\frac{u_i^{n+1}}{\Delta t} - \frac{u_i^n}{\Delta t}}_{\partial_t u + O(\Delta t)} = -a \underbrace{\left(\quad \right)}_{\partial_x u + O(\Delta x^2)}$$

$$u_i^{n+1} = u_i^n - a\Delta t \underbrace{(\partial_x u)_i^n}_{\text{2nd ORDINE SPAZIO}} + \frac{\Delta t^2}{2} a^2 \underbrace{(\partial_{xx} u)_i^n}_{\text{1° ORDINE}}$$



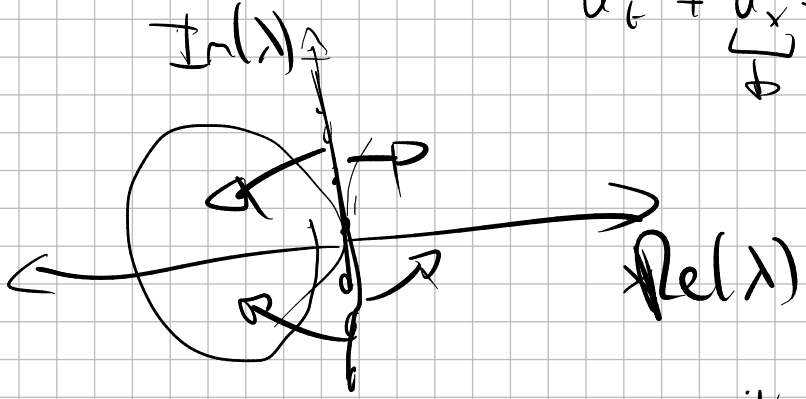
$$\frac{u_{i-2} - 4u_{i-1} + 3u_i}{2\Delta x} \qquad \frac{u_{i-2} - 2u_{i-1} + u_i}{2\Delta x^2}$$

È STABILE? SÌ

FISSO A CFL ≤ 2

$$\partial_x \left(e^{ixk}\right) = \underbrace{ik}e^{ikx}$$

$$u_t + \underbrace{u_x}_{} = 0$$



$$\underbrace{\partial_{xx} u} = \partial_{xx} e^{ikx} = \underbrace{-k^2} e^{ikx}$$

BORDO $\nearrow$ $a>0$

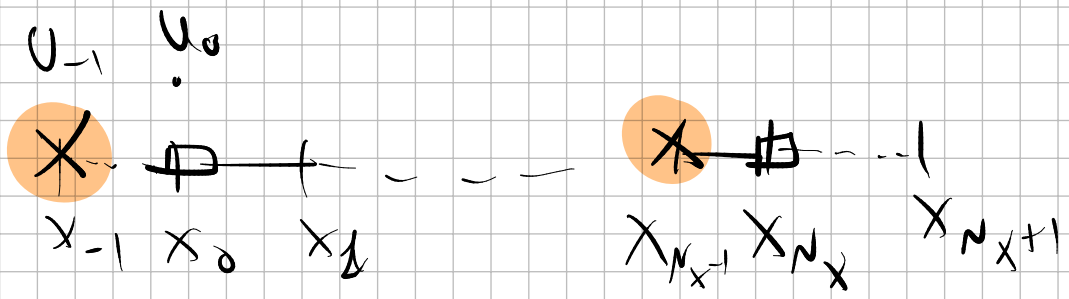
- DIRICHLET $\Rightarrow$ solo A SX se $a>0$

  solo A DX se $a<0$

- NEUMANN $a>0$ "FAR FINTA" CHE NEUMALU A DX

  $\downarrow$ NO PER I PROBLEMI

GHOST POINT



DIRICHLET es SSX $a>0$

$$u(x_{-1}) = \underline{U_D}$$

OUTFLOW es A D X $a>0$

$$u(x_{N_x+1}) = U_{N_x}$$

PERIODICO es a SX

$$U(x_{-1}) = U_{N_x-1}$$